

- Q.23 What are colloidal impurities? How these affect the water? (CO3)
- Q.24 Write a short note on break-point chlorination. (CO4)
- Q.25 What is meant by screening? Explain its various types. (CO4)
- Q.26 Write a short note on reflux valve. (CO5)
- Q.27 Write a short note on testing of water pipe lines. (CO6)
- Q.28 Explain combined sewerage system. (CO9)
- Q.29 Explain various methods of sanitation. (CO8)
- Q.30 What are the objectives of sewage disposal? (CO8)
- Q.31 What do you mean by storm sewage? Differentiate sanitary sewage with storm sewage. (CO9)
- Q.32 Explain the concept of B.O.D and its importance. (CO11)
- Q.33 What are the method of land treatment? Explain in brief. (CO12)
- Q.34 Write a short note on sludge disposal. (CO13)
- Q.35 Write various sanitary fittings. (CO14)

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What do you mean by sewer appurtenances? What are the different Appurtenances used for sewage system. (CO9)
- Q.37 Explain self purification of streams. (CO12)
- Q.38 Explain layout of water distribution system. (Co6)

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#### 4th Sem / Civil, Brick Tech, Constr. Mgmt. Highway Engg.

#### Subject:- Water Supply and Waste Water Engineering / Public Health Engg.

Time : 3Hrs. M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Design period for water supply is \_\_\_\_\_ years. (CO2)
- a) 40-50 years b) 20-30 years  
c) 1015 years d) 30-40 years
- Q.2 Water demand of a city includes (CO2)
- a) Domestic water demand  
b) Commercial and industrial demand  
c) Fire and public use demand  
d) All of these
- Q.3 PH value representing a stronger acid is (CO3)
- a) 1 b) 5  
c) 7 d) 10
- Q.4 The first stage in water treatment is (CO4)
- a) Sedimentation  
b) Filtration  
c) Screening  
d) Coagulation and mixing

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- Q.5 Reflux valve is also called \_\_\_\_\_ (CO5)  
 a) Pressure relief valve  
 b) Check valve  
 c) Stop valve  
 d) Scour valve
- Q.6 The type of joint generally used in cast iron pipes is \_\_\_\_\_ (CO5)  
 a) Flanges joint  
 b) Socket and spigot joint  
 c) Dresser coupling joint  
 d) All of these
- Q.7 The normal flow in a sewer during dry weather is known as \_\_\_\_\_ (CO8)  
 a) D.W.F                      b) W.W.F  
 c) Drainage                  d) Sewerage
- Q.8 Sanitary sewage includes \_\_\_\_\_ (CO9)  
 a) Domestic sewage      b) Industrial sewage  
 c) Both (a) and (b)      d) None of these
- Q.9 For joining the sewers at different levels, the device or appliances used is called \_\_\_\_\_ (CO5)  
 a) Drop manholes      b) Man-holes  
 c) Both (a) and (b)      d) None of these
- Q.10 A long basin constructed for the purpose of removing grit, sand etc from sewage is known as \_\_\_\_\_ (CO13)  
 a) Skimming tanks      b) Grit chamber  
 c) Screening              d) Coagulation

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## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Per capita consumption for domestic use is \_\_\_\_\_. (CO1)
- Q.12 PH value of water should be \_\_\_\_\_ to \_\_\_\_\_. (CO3)
- Q.13 \_\_\_\_\_ are the factors responsible for the change in population. (CO2)
- Q.14 Slow sand filters require \_\_\_\_\_ area. (CO4)
- Q.15 The process of preventing the entry or large sized particles such As dead animals, branches into intake words etc is known as \_\_\_\_\_. (CO2)
- Q.16 Dual system means \_\_\_\_\_. (CO2)
- Q.17 Kitchen waste is \_\_\_\_\_. (CO8)
- Q.18 The natural process in which the oxygen of water of streams is consumed by Sewage and at the same time, it is replenished by the atmosphere is know as \_\_\_\_\_. (CO12)
- Q.19 Aerobic bacteria exists in the presence of \_\_\_\_\_. (CO13)
- Q.20 The drainage pipes should be made of \_\_\_\_\_. (CO14)

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on variations in water demand. (CO2)
- Q.22 What is population forecasting? Explain incremental increase method. (CO2)

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